

Data-edge batching in the two-stage interchange

Draft

Setting. The two-stage bit motif with partial-swap frames, $m = h^2$, $v = \binom{h}{3}$, $N = v^2$, with every auxiliary role's selected edge compiled by the partial-swap batching lemma (`partial-swap-batching.tex`). Write P_a for the G -orthogonal projector onto $\langle t_a \rangle$ and $Q_a = I - P_a$.

Lemma 1 (Data entrances). *For every data pair, the stage-two entrance edges of the X and Y roles both have idempotent difference $Q_1 \otimes Q_2$, of rank $(h-1)^2 = m - 2h + 1$.*

Proof. Since $t_1^\perp \otimes F \perp t_1 \otimes t_2$ under $G \otimes G$, the X entrance is $\text{proj}(D_0 + t_1 \otimes t_2) - \text{proj}(F \otimes t_2) = Q_1 \otimes I + P_1 \otimes P_2 - I \otimes P_2 = Q_1 \otimes Q_2$. The Y entrance is $\text{proj}(D_0) - \text{proj}(t_1^\perp \otimes t_2) = Q_1 \otimes I - Q_1 \otimes P_2 = Q_1 \otimes Q_2$. $\square$

For $h \geq 4$, $(h-1)^2 > m/2$, so the partial-swap batching lemma applies with $k = 2h - 1$ corner transpositions and one aligned block of width $t = m - 4h + 2$. The common basis is the one already used for the auxiliary edges: each data corner adds one nonzero polynomial condition $\det(S^{-1}U)_L \det(VS)_R$ on S , nonzero on the big cell $S = LU$ (unit lower times unit upper), so finitely many conditions have a common integral unimodular solution.

Accounting. Per data role the ranks are $(h-1) + (m - 2h + 1) + (h-1) = m - 1$; with one copy-correction child of width 1 per pair this reproduces $s = Wm - N + 2L$, and $Wm - s = v(v - 4h^2) > 0$ for $h \geq 28$.

Saving. With the side-role counts of our generator (valid for $28 \leq h \leq 50$), the batched moment certifies $a_b = 10033/10^9$ at $h = 47$ ($R = 418678$), against $10027/10^9$ at $h = 46$ and $4431/10^9$ at $h = 32$ without data batching.

Checks. Corners invertible under one common S_0 for every auxiliary edge and every data pair at $h = 6, 7, 8$ (all 3136 pairs at $h = 8$); exact Bruhat profiles match; the closed form is checked in rationals.